



The influence of Electronic Supervision Practice on Supervisory Performance among Supervisors of Ministry of Education in the Sultanate of Oman

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Abstract:

This study aimed to evaluate the effectiveness of electronic supervision and its relationship with supervisory performance among supervisors at the Ministry of Education in the Sultanate of Oman. A total of 447 supervisors participated in the study. The research employed utilized descriptive and correlational research methods, including questionnaires and statistical analyses such as t-tests and ANOVA. The findings revealed that supervisors demonstrated high competence in electronic supervision methods ($M = 4.07$, $SD = 0.55$) and applications ($M = 3.96$, $SD = 0.67$), with an overall mean score for e-supervision of $M = 3.74$ ($SD = 0.59$). However, their overall electronic supervision competencies were rated as moderate ($M = 3.35$, $SD = 0.76$). Positive correlations were found between electronic supervision practices and various dimensions of supervisory performance, with the strongest correlation observed between electronic supervision methods and supervisory performance ($r = 0.632$, $p < 0.01$). Additionally, gender differences were examined. The results indicated no significant differences in most dimensions of electronic supervision and supervisory performance, except in professional development, where female supervisors scored higher than male ($M = 4.40$, $SD = 0.53$ for females; $M = 4.29$, $SD = 0.51$ for males, $t(445) = -2.322$, $p = 0.021$). Furthermore, years of experience did not show any significant differences in electronic supervision or supervisory performance ($p > 0.05$). Based on these findings, the study recommends enhancing training programs, particularly in areas where supervisors demonstrated moderate competence, and promoting continuous professional development for all supervisors regardless of gender or

experience level. These efforts aim to improve the overall effectiveness of electronic supervision in the Omani education sector.

Keywords: electronic supervision, supervisory performance, relationship, gender differences, years of experience.

Introduction

Educational supervision employs a variety of methods within the supervisory process, including guided readings, exchange visits, meetings, training programs, educational workshops, lectures, electronic supervision, and others. These approaches aim to enhance the supervisory performance of educational supervisors and improve their professional competencies. The quality of supervisory performance is closely linked to the diversity of supervisory services provided, all of which aim to advancing the educational process in its entirety. This approach aligns with modern trends in educational supervision, which focus on key elements such as the teacher, curriculum, student, the internal environment (the school), the external environment (the local community), and the importance of human relationships among colleagues and with the local community (Al-Kiyumi and Hammad, 2020).

As educational supervision plays a critical role in monitoring the teaching and learning process, the integration of modern technology has led to a shift from traditional supervisory practices toward more dynamic approaches capable of addressing contemporary challenges. Consequently, educational supervision has adopted new terminology, including "electronic supervision" and "remote supervision," reflecting its technical and digital orientation. These innovations have played a crucial role in overcoming the material and administrative challenges that educational supervision traditionally faced (Astuti, Sutiah, Hidayatulloh, and Nisak, 2024).

Electronic supervision, as a modern trend in educational supervision, relies on the use of electronic tools and contemporary technologies to facilitate communication, whether synchronous or asynchronous. This method provides continuous access to supervisory resources, including supervisory methods, information, readings, and model lessons, which are readily available always via websites. It contributes to cognitive development, supports, ongoing pretraining, and helps overcome geographical and temporal barriers (Ugwoke, 2011).

In the context of educational guidance, Ahmed (2021) highlighted that the effectiveness of educational counselors in fulfilling their advisory roles—especially in educational settings—depends on both internal factors such as the counselor's personal attributes, skills, professional attitudes, readiness, and motivation, as well as external factors such as training, preparation, work environment, and supervision. Rahyasih, Syarifah, & Syarifah (2024) emphasized that the concept of electronic supervision extends beyond the use of technical tools and devices. It represents an integration of various forms of supervision, such as open supervision, internet-based supervision, self-supervision, correspondence supervision, postal supervision, and remote supervision. These approaches are designed to enhance the performance and professional development of both supervisors and teachers.

The primary objective of electronic supervision is to improve the methods used in educational supervision. It is a flexible and adaptable form of supervision that facilitates faster decision-making and relies on electronic documentation, thus reducing costs and improving efficiency. electronic supervision also ensures the continuous monitoring of the educational processes by making supervisors' work more efficient, enhancing communication between all elements involved in the supervisory process, reducing the stress typically associated with supervisory

visits, and maximizing the potential of various supervisory methods (Mahmudi, Ulfa, & Rofiq, 2024)

Fatimah (2022) explained that the rise of electronic supervision was driven by the need for knowledge exchange and access to diverse sources of information to foster self-professional development and achieve self-improvement among all participants in the supervisory process. Considering the challenges faced by supervisors, it has become essential to focus on improving their professional performance, enabling them to face future challenges related to technological, informational, and cognitive advancements. Vaiz, Minalay, Türe, Ülgener, Yaşar, & Bilir, (2021), emphasized the necessity of continuous and ongoing evaluation of the supervisory process to assess the development of supervisors' performance and determine their scientific and professional competency. This need has underscored the importance of performance indicators as tools for measuring effectiveness and efficiency.

In this context, the present study aims to explore the practices of e-supervision among supervisors at the Ministry of Education in the Sultanate of Oman and evaluate its impact on supervisory performance.

Research problem

The supervisory process in educational institutions under the Ministry of Education in the Sultanate of Oman faces several challenges, including a substantial shortage of supervisors in educational directorates compared to the number of teachers and school administrations. Additionally, supervisors face significant administrative and technical burdens, compounded by the geographic distance between the educational directorates and schools. This situation is corroborated by the study of Al-Zeidi, Al-Zoubi, and Al-Kiyumi (2021), which identifies several difficulties within the supervisory framework in Oman, such as long travel distances, a lack of supervisors, limited integration of

technology during supervisory visits, an emphasis on theoretical aspects of visits, and the growing workload placed on supervisors.

The findings of numerous previous studies underscore the necessity of integrating technology and electronic applications into educational supervision. For example, Al-Deek (2023) recommended enhancing the role of social media in the supervisory process by encouraging educational supervisors to create groups for the schools and teachers they oversee, enabling communication and providing them with links to educational and instructional websites. This recommendation is further supported by the studies of Ohara (2023) and Salem (2020), which highlight the importance of senior and middle leadership in monitoring and encouraging educational supervisors. Their studies emphasize the need for supervisors to adopt modern technology, including opening email accounts, utilizing computers, and leveraging the internet in their supervisory methods. Furthermore, they advocate for a shift from traditional supervision to electronic supervision through the activation of social media and the allocation of necessary funding to support the infrastructure for electronic supervision.

In light of the importance of technological advancements in enhancing supervisory practices and the Ministry of Education's commitment to integrating technology into both supervisory and educational processes, in alignment with Oman Vision 2040, this study aims to explore the relationship between e-supervision and supervisory performance among supervisors.

Research questions

The current study seeks to answer the following questions:

1. What is the level of electronic supervision among supervisors at the Ministry of Education in the Sultanate of Oman?

2. What is the relationship between electronic supervision and supervisory performance among supervisors at the Ministry of Education in the Sultanate of Oman?
3. Are there any significant differences between electronic supervision and supervisory performance among supervisors at the Ministry of Education in the Sultanate of Oman based on gender and years of experience?

Definition of Keyword

E-supervision

E-supervision is defined by Habibi et al. (2020) as a supervisory model that delivers the supervisor's services to teachers and educational institutions through multimedia, enabling active interaction.

The researcher operationally defines E-supervision as the supervisory practices conducted by educational and school administration supervisors in the Sultanate of Oman, always utilizing modern technological tools and electronic applications and across various locations, with the aim of achieving the objectives of the supervisory process. It is measured based on the responses of the study sample to the research instrument.

Supervisory Performance

Ampofo et al. (2020) defines supervisory performance as the execution of various tasks and activities that constitute an individual's work within a specified time frame. It also refers to the extent to which the educational supervisors fulfill their assigned duties, including the practices, activities, and behaviors they engage in related to these responsibilities, as manifested behaviorally.

In operational terms, the researcher defines supervisory performance as the efforts, activities, and supervisory practices undertaken by supervisors to enhance their knowledge, skills, and behaviors. Thus, enabling supervisors to perform their roles effectively and achieve the desired outcomes in improving the

supervisory process. The efforts include participation in training programs, educational workshops, exchange visits, and other supervisory strategies. As well as the integration of technological tools in their supervisory tasks. The measurement of these practices is based on the responses of the sample to the instrument related to supervisory performance used in this study.

Review of Related Literature

Supervisory Performance

Educational supervision is regarded as a crucial and effective tool for enhancing the quality of education in schools. The role of educational supervision involves providing support and guidance to teachers, ensuring the implementation of effective teaching practices, and achieving successful learning outcomes for students. One of the primary objectives of educational supervision is to foster the development of teaching practices and improve the overall educational process within schools. Supervisors offer guidance to teachers by observing lesson preparation, monitoring, and analyzing its implementation. Supervisors also provide constructive feedback and recommendations to enhance teaching methods and the integration of modern educational technologies. This support helps to boost teachers' efficiency and develop their educational competencies. Performance in educational supervision is a fundamental pillar that contributes to improving the effectiveness of the educational process and enhancing its outcomes. The educational supervisor plays a central role in achieving educational objectives by guiding and supporting teachers, which, in turn, improves the learning environment and raises the quality of education. The success of educational supervision is largely contingent upon the supervisor's ability to assess performance and continuously direct teachers toward achieving the established educational goals (Hepriyanti, 2023).

One of the most critical factors contributing to improving performance in educational supervision is ongoing training for both supervisors and teachers. Training helps to develop the skills of teachers and supervisors in various areas, such as classroom leadership, student interaction, and the application of modern technological tools in teaching. Continuous training is considered an essential tool for elevating overall educational performance, as it enhances the supervisor's ability to provide a better and more innovative learning environment. In addition to training, performance evaluation plays a vital role in educational supervision. Supervisors must be able to establish clear performance standards, which include aspects such as teaching effectiveness, teacher-student interaction, and the quality of instructional methods employed. Continuous evaluation highlights both strengths and weaknesses in teachers' performance, contributing to the provision of necessary support for skill enhancement. Furthermore, evaluation allows supervisors to adjust their supervisory strategies to ensure that educational objectives are effectively met (Catherine & Andala, 2024).

Effective communication between supervisors and teachers is another essential factor in improving performance in educational supervision. Open and ongoing communication helps build a cooperative and trusting relationship between the two parties, enhancing teachers' understanding of students' needs and curriculum requirements. It also allows supervisors to provide constructive feedback that contributes to the improvement of teaching methods and motivates teachers to continually enhance their performance. Despite the many benefits, supervisors in educational supervision face various challenges, such as managing the diversity of teaching methods and addressing the rapid changes in educational curricula. It is essential for educational supervisors to adapt to these challenges

and guide teachers toward innovative and effective teaching strategies that cater to the diverse needs of students. Additionally, supervisors must provide both moral and material support to teachers, which boosts their motivation and encourages them to continue improving their performance.

In conclusion, performance in educational supervision is a critical element in enhancing the quality of education and developing the educational environment. Through continuous training, effective evaluation, and strong communication, educational supervisors can make a significant impact on improving teachers' performance, thereby enhancing overall educational outcomes (Alkhazaleh & Hattamleh, 2019).

Electronic Supervision

E-supervision is an innovative model of educational supervision that involves delivering supervisory tasks and activities through multimedia platforms on computers and their networks. This approach allows teachers and schools to actively engage with educational supervisors or peers, either synchronously or asynchronously, offering flexibility in completing these processes at times and locations that best suit the supervisors' circumstances. Moreover, the management of these processes is streamlined using such multimedia platforms.

E-supervision is characterized by several key features. It presents educational supervision tasks in a digital multimedia format, which may include written or spoken texts, sound effects, graphics, animations, static images, and video clips. These tasks are facilitated through software on personal computers, local networks, or the internet, enabling interactive engagement between teachers and supervisors. The model supports synchronous interaction, where communication occurs in real-time through chat rooms, video conferences, or audio calls, as well as asynchronous interaction, where communication happens at different times via email or forums. This flexibility allows

teachers to engage in learning at their own convenience, significantly improving accessibility and convenience (Kaomaroeng & Rattanavaraha, 2024).

The need for e-supervision stems from its potential to provide effective solutions to the challenges which has faced in educational supervision, especially within the broader educational context. Upon reviewing the current supervisory system, it is evident that there is an urgent need to adopt e-supervision for several reasons. The existing supervisory framework struggles to keep pace with the rapid growth in information volume and diversity. It is also insufficient to meet the growing demand for monitoring schools and enhancing staff performance. Furthermore, the number of schools and teachers continues to grow, whereas the number of supervisors grows at a slower rate. Current training institutions are unable to meet the ongoing demand for specialized training for supervisors, principals, and teachers. Additionally, teacher preparation institutions failed to produce or retrained highly qualified supervisors and teachers to meet the evolving educational needs (Haleem et al., 2022).

E-supervision aims to achieve the core objectives of educational supervision, though it utilizes different methods and techniques compared to traditional practices. The model seeks to: introduce educators to new pedagogical methods, modern trends in management, curricula, teaching tools, and assessment strategies through software; enhance professional competence and foster self-directed professional development; respect the individuality of each participant, empowering them to identify and analyze problems independently; facilitate communication between educational authorities, schools, and the local community through electronic connections; and strengthen relationships and collaboration among educators, as well as between teachers and their peers (Ismail, 2018).

E-supervision offers several advantages that contribute to professional development and improve the educational process. It enables teachers to engage in professional growth anytime and anywhere, facilitating training and continuous professional development within their work environments without requiring them to leave their duties or find substitutes. The model encourages communication and collaboration among teachers, creating a supportive professional network. Remote supervision addresses issues related to overcrowded or unavailable training facilities. Additionally, it helps both supervisors and teachers to stay updated with the rapid changes in educational knowledge and practices. It enhances the ability of supervisors to effectively communicate with schools and access information efficiently. E-supervision also addresses performance and qualification gaps among some educational supervisors, reduces the tendency for inspection-based practices, and contributes to improving overall educational quality (Astuti et al., 2024).

Despite its numerous benefits, e-supervision faces several challenges that need to be addressed. These include a lack of adequate infrastructure, such as devices, networks, and modern communication methods, to support this model of supervision. There is also a shortage of qualified human resources, including both technical staff (e.g., software developers, administrators) and educational personnel (e.g., supervisors, principals, and teachers). Additionally, many supervisors, principals, and teachers possess limited skills in computer usage and internet navigation. Moreover, the high financial costs associated with implementing e-supervision, including the purchase of necessary devices, software, and internet connectivity, pose another challenge. Resistance from supervisors, principals, and teachers towards adopting e-supervision practices is also a significant barrier. Finally, transitioning from traditional views of educational supervision, which focus primarily on physical

visits, directive approaches, and performance evaluations, to more modern, digital methods is a difficult process (Taole et al., 2024).

Previous studies

The study by Kaomaroeng et al. (2024) aimed to verify the effectiveness of digital educational supervision provided by educational supervisors affiliated with the Education Service Office in the Northeastern Region of Thailand. The study relied on a developmental approach that included three stages: investigating the effectiveness of digital educational supervision by reviewing relevant literature and documents, conducting interviews with (30) participants with outstanding performance, developing a prototype educational supervision model and evaluating it by (7) experts using the critical evaluation method, and evaluating the feasibility and benefits of the model using questionnaires on a sample of (50) educational supervisors. The results showed that the developed model consists of five main components: principles, objectives, basic content, implementation guidelines, and success conditions, and achieved a very high level of feasibility and benefit according to the sample evaluation.

The study by Mahmudi, Ulfa, & Rofiq (2024). aimed to optimize the role of supervision in the era of technology based on digital learning. The object of this research is Technology-Based Supervision: Optimizing the Role of Supervision in the Digital Learning Era at SMA Darussalam Blokagung Banyuwangi East Java. In this research, the method used is qualitative with a case study approach to examine the effectiveness of technology-based supervision at SMA Darussalam Blokagung, Banyuwangi, East Java. The source of informants in this study consisted of the Quality Control Team (TPM), the principal, and teachers at SMA Darussalam Blokagung Banyuwangi, East Java. Data collection techniques in

this study include observation, interviews, and documentation, which aimed to gain a comprehensive understanding of the effectiveness of technology-based supervision at SMA Darussalam Blokagung Banyuwangi, East Java. Data analysis in this study used the data analysis model proposed by Miles and Huberman, which consists of three main steps: data reduction, data presentation, and conclusion drawing. The results of this study are the increased effectiveness of supervision through digital tools, the ability of supervisors to monitor the development of learning innovations globally and the improvement of teacher competence in technology through digital supervision.

The study by Ahmed (2021) aimed to determine the effectiveness of electronic supervision for supervisee groups in the school field. This study applied the descriptive and analytical approaches; it used social survey as a method for collecting social data through questionnaires. Moreover, it was applied to social workers working in school within Muscat Governorate of Oman, during the period from September 2020 until November 2020. The results showed the importance of electronic supervision and indicated a variety of roles for electronic supervision in the school field. Finally, this study demonstrates that electronic supervision must be activated with supervisee groups in the school field.

The study by Habibi et al. (2020) aimed to analyse in depth the perceptions of teachers in Bukit Tinggi city towards e-supervision using the web. This type of research is descriptive research. The sample of this study was all teachers at SD Al-Azhar Bukit Tinggi, which consisted of 28 people. The study used an interview and questionnaire tool. The results showed that 75% of teachers responded to the development of electronic supervision, as electronic supervision can be carried out using the web anywhere and at any time, and it is easier for teachers to

interact and discuss with the principal and colleagues about the problems they face in the learning process. Electronic supervision is a good way to avoid miscommunication between principals and teachers in communicating supervision results, and teachers will obtain clear supervision results due to this systematic supervision integrated with technology.

The study by Gumbo (2019) inquired into the supervisors' knowledge of the technological tools and their stance in the use of (online) technology in their supervision given the human aspect. The research involved 129 supervisors from the College of Education (CEDU) at UNISA, focusing on their supervisory characteristics, exposure to technology, and familiarity with technological tools. This was complemented by face-to-face interviews with seven supervisors to assess their views on integrating technology with the human aspect of student supervision. The findings revealed that the supervisors possessed a good understanding of various tools and applications, such as MS Word, my Unisa, and Discussion Forum. Additionally, the study highlighted the need for training other supervisors in using specific online applications and tools. Most importantly, the results indicated that supervisors highly value the human aspect. The study by Vaiz et al. (2021) examines the concept of supervision in the distance education process and the emerging e-supervision concept. In the study conducted with the document analysis method, which is one of the qualitative research methods, the relevant literature was examined, and the findings obtained were discussed. As a result of the research, it has been revealed that the advantages of distance education also provide opportunities for e-supervision to facilitate control, eliminate time and place concepts, and create an easier communication network between supervisors, teachers and school administrators.

Research Methodology

In this study, the researcher employed a quantitative approach. Quantitative research is a systematic method that posits the existence of objective social facts, separate from the emotions and beliefs of individuals. This approach mainly depends on statistical methods for both data collection and analysis, encompassing both descriptive and correlational studies. As such, the current study applies a correlational-causal methodology to examine the quantitative data. This approach is descriptive in nature and aims to explore the relationships between variables.

Research population and sample.

This study examines a population of 1,408 supervisors from the General Directorate of Educational Supervision and the educational supervision departments across the educational directorates. The sample comprises 447 supervisors, with 271 males (60.6%) and 176 females (39.4%). In terms of experience, 32% have 10 years or less, 37.2% have 11-20 years, 27.5% have 21-30 years, and 3.4% have more than 31 years. Job titles within the sample include 4.5% as department heads, 19.9% as senior school supervisors, and 75.6% as senior educational or subject supervisors. The sample is distributed across various educational directorates in Oman.

Research instrument

The research tool employed in this study is divided into two main sections. The first section covers the demographic characteristics of the study sample, such as gender, supervisory experience, educational qualifications, and the educational directorate. The second section examines the two key study variables, each with its own set of dimensions.

The first variable, Electronic Supervision, which includes five main areas: Electronic Supervision Models and Patterns with 10 items, Electronic Supervision Competencies with 12

items, Electronic Supervision Methods with 9 items, Electronic Supervision Applications with 14 items, and the Educational Indicators System with 15 items. Therefore, the total number of items in the Electronic Supervision domain is 60 items. The second variable, Supervisory Performance, includes six dimensions: Supervisory Planning (10 items), Curriculum Development and Teaching Methods (10 items), Professional Development (10 items), Human Relations (11 items), Performance Evaluation (10 items), and Administrative and Technical Performance (15 items), totaling 66 items.

First: content validity

The researcher ensured the validity of the questionnaire by presenting its initial version to the thesis supervisor for feedback and subsequent modifications. After that, it was reviewed by 22 experts, including faculty members from Sultan Qaboos University, Nizwa University, Sohar University, the Arab Open University, and specialists from the Ministry of Education in Oman. These experts were carefully selected for their expertise in research, supervision, school management, and language formulation. The researcher asked them to evaluate the clarity, appropriateness, and relevance of the items to ensure they accurately measured the intended constructs and aligned with the study's objectives.

Following the expert reviews, the researcher gathered all feedback and adjusted accordingly. In the Electronic Supervision questionnaire: the number of removed items is 37, the number of modified items is 23, and no new items were added by the reviewers. As a result, the final number of items for this domain is 23. Additionally, two dimensions were removed from this variable based on the reviewers' comments, as these dimensions were found to overlap with each other. The remaining dimensions are: Electronic Supervision Competencies, Electronic Supervision Methods, and Electronic Supervision

Applications. In the supervisory performance questionnaire, 36 items were deleted, and 30 were reworded, leaving a total of 30 items in the final version. These changes were made to enhance the validity and relevance of the study tool.

Second: Construct Validity

The researcher evaluated the Construct validity of the study's questionnaire by assessing its discriminatory power using the extreme groups method. This procedure involved selecting 33% of participants with the highest and lowest scores from a sample of 40 individuals, and their responses were compared using a T-test. The findings revealed statistically significant differences ($p < 0.05$) between the high and low score groups across the two scales: e-supervision and supervisory performance. Specifically, the T-values for each scale were significant, demonstrating that the scales effectively distinguish between participants with high and low scores. These results confirm the strong discriminatory power of the scales and further validate the overall effectiveness of the questionnaire.

Third: Reliability

The researcher calculated Cronbach's Alpha coefficients to evaluate the reliability of two variables and their respective dimensions through a pilot study involving 40 participants. As presented in Table 1, the Cronbach's Alpha values for the study's scales demonstrated high reliability across all areas. For E-supervision, the values were as follows: Electronic Supervision Competencies: 0.859, Electronic Supervision Methods: 0.919, Electronic Supervision Applications: 0.786, and total E-supervision: 0.942. For supervisory performance, the values were: Supervisory Planning: 0.736, Curriculum Development and Teaching Methods: 0.850, Professional Development: 0.790, Human Relations: 0.899, Performance Evaluation: 0.856, Administrative and Technical Performance: 0.836, and total Supervisory Performance: 0.939.

Table (1): Cronbach's Alpha Coefficients for Study variables and Their Dimensions

Dimensions	Cronbach's Alpha Coefficients
E-supervision	
Electronic Supervision Competencies	0.859
Electronic Supervision Methods	0.919
Electronic Supervision Applications	0.786
Total for E-supervision	0.942
Supervisory Performance	
Supervisory Planning	0.736
Curriculum Development and Teaching Methods	0.850
Professional Development	0.790
Human Relations	0.899
Performance Evaluation	0.856
Administrative and Technical Performance	0.836
Total for Supervisory Performance	0.939

RESULTS AND DISCUSSION

Q1. What is the level of Electronic Supervision among supervisors at the Ministry of Education in the Sultanate of Oman?

To find the level of Electronic Supervision among supervisors at the Ministry of Education in the Sultanate of Oman, the researcher used the Weighted average interpretations of five-point Likert scale results (see table (2)).

Table (2) Weighted Averages for 5-Point Likert Scale

Weighted Average	Result
1-1.79	Very low
1.80-2.59	Low
2.60-3.39	Moderate
3.40-4.19	High
4.20-5	Very high

Table (3) presents the means and standard deviations for various dimensions of E-supervision among supervisors at the Ministry of Education in the Sultanate of Oman. Starting with Electronic Supervision Competencies, the mean was observed to be

moderate ($M = 3.35$, $SD = 0.76$), indicating that participants possess a moderate level of competence in electronic supervision. This was followed by Electronic Supervision Methods ($M = 4.07$, $SD = 0.55$) and Electronic Supervision Applications ($M = 3.96$, $SD = 0.67$), both of which were rated as high, reflecting strong perceptions of competence in applying electronic methods and applications of supervision.

The total mean for E-supervision was also rated high ($M = 3.74$, $SD = 0.59$), suggesting that, overall, participants feel relatively confident in their ability to carry out electronic supervision tasks. The results indicate that while supervisors show high confidence in certain aspects of electronic supervision, such as methods and applications, there is room for improvement in terms of their overall competence in this area.

Table (3) Means and standard deviation of E-supervision among supervisors at the Ministry of Education in the Sultanate of Oman

Dimension	Mean	Std. Deviation	Decision
Electronic Supervision Competencies	3.35	0.76	Moderate
Electronic Supervision Methods	4.07	0.55	High
Electronic Supervision Applications	3.96	0.67	High
Total for E-supervision	3.74	0.59	High

This result is consistent with the findings of study by Gumbo (2019), which indicated findings showed that the supervisors are knowledgeable in terms of the tools or applications such as MS Word, my Unisa and Discussion Forum. Furthermore, the findings indicated that there is a need for the training of other supervisors with respect to the use of certain online applications and/or tools. It also aligns with the study by Habibi et al. (2020), which found that 75% of teachers positively responded to the development of this e-supervision using web. Additionally, the results support the findings of Vaiz et al. (2021) whose study revealed that the advantages of distance education also provide opportunities for e-supervision to

facilitate control, eliminate time and place concepts, and create an easier communication network between supervisors, teachers and school administrators.

Q2 What is the relationship between Electronic Supervision and supervisory performance among supervisors at the Ministry of Education in the Sultanate of Oman??

The table (4) presents the correlations between different aspects of Electronic Supervision and various dimensions of Supervisory Performance. The correlations are generally positive and significant across all areas. Electronic Supervision Competencies show moderate positive correlations with Supervisory Planning (0.472**), Curriculum Development and Teaching Methods (0.463**), and Supervisory Performance (0.510**). Electronic Supervision Methods exhibit stronger correlations, with Supervisory Performance (0.632**) showing the highest value, followed by Performance Evaluation (0.577**) and Human Relations (0.478**). Similarly, Electronic Supervision Applications show positive correlations with all aspects of supervisory performance, with Supervisory Performance (0.620**) and Professional Development (0.560**) standing out.

However, there are also lower correlations in some areas. Human Relations have the weakest correlation with Electronic Supervision Competencies (0.273**) compared to other dimensions of supervisory performance. Additionally, the correlation between Curriculum Development and Teaching Methods and Electronic Supervision Methods (0.391**) is also relatively low. The total for E-supervision displays moderate to strong correlations, with the highest being with Supervisory Performance (0.632**) and Performance Evaluation (0.543**), further indicating that electronic supervision is positively associated with effective supervisory performance, particularly in evaluation and planning processes.

Table (4): Correlation Coefficients of the Relationship Between Electronic Supervision and Supervisory Performance

	Supervisory Planning	Curriculum Development and Teaching Methods	Professional Development	Human Relations	Performance Evaluation	Administrative and Technical Performance	Supervisory Performance
Electronic Supervision Competencies	.472**	.463**	.455**	.273**	.414**	.373**	.510**
Electronic Supervision Methods	.593**	.391**	.542**	.478**	.577**	.527**	.632**
Electronic Supervision Applications	.547**	.437**	.560**	.454**	.541**	.495**	.620**
Total for E-supervision	.582**	.483**	.557**	.412**	.543**	.494**	.632**

** Correlation is significant at the 0.01 level (2-tailed).

This result is consistent with the findings of the study by Mahmudi, Ulfa, & Rofiq (2024), which indicated the increased effectiveness of supervision through digital tools, the ability of supervisors to monitor the development of learning innovations globally and the improvement of teacher competence in technology through digital supervision. It also aligns with the study by Quzuq (2022), which found that the sample demonstrated a "Good" level of academic self-efficacy. Both studies suggest a positive, though moderate, perception of self-efficacy and related competencies, similar to the present study where participants exhibited strong confidence in their individual, collective, cognitive, and emotional self-efficacy. However, there was slightly more variation in areas such as organizational self-efficacy and supervisory skills, reflecting some aspects of previous research. Additionally, the results support the findings of Muna Al Aalwi (2022), whose study revealed that there was statistically significant relationship at ($\alpha \leq 0.01$) between the educational supervision and electronic leadership and its obstacles in public schools in the northern West Bank governorates as perceived by the educational

supervisors, as the values of Pearson correlation were (0.52, 0.35) respectively.

Q3 Are there significant differences in Electronic Supervision and Supervisory Performance among supervisors at the Ministry of Education in the Sultanate of Oman based on gender and years of experience?

The researcher used two independent sample t-tests independent sample two t-tests to examine the differences based on gender and applied a one-way ANOVA to assess the differences based on years of experience. Table 5 presents the results of independent samples t-tests conducted to examine the differences in Electronic Supervision and Supervisory Performance based on gender.

For Electronic Supervision Competencies, the mean scores for males ($M = 3.37$, $SD = 0.78$) and females ($M = 3.31$, $SD = 0.73$) were relatively close, $t(445) = 0.777$, $p = 0.438$, indicating no significant difference between the genders. Similarly, for Electronic Supervision Methods, males had a mean score of 4.04 ($SD = 0.55$), and females had a mean of 4.11 ($SD = 0.55$), $t(445) = -1.23$, $p = 0.22$, suggesting no significant gender differences. The scores for Electronic Supervision Applications were identical for both genders ($M = 3.96$, $SD = 0.66$ for males, $M = 3.96$, $SD = 0.67$ for females), $t(445) = -0.092$, $p = 0.927$, indicating no difference. The Total for E-supervision also showed no gender-based differences, as the means for both males ($M = 3.74$, $SD = 0.60$) and females ($M = 3.74$, $SD = 0.58$) were identical, $t(445) = -0.036$, $p = 0.971$.

For Supervisory Planning, males ($M = 4.26$, $SD = 0.53$) and females ($M = 4.32$, $SD = 0.53$) showed a t-value of -1.187, $p = 0.236$, indicating no significant difference. Similarly, Curriculum Development and Teaching Methods showed no significant difference, $t(445) = -0.65$, $p = 0.516$. However, a significant difference was found for Professional Development,

where males had a mean score of 4.29 (SD = 0.51) and females had a mean score of 4.40 (SD = 0.53), $t(445) = -2.322$, $p = 0.021$, indicating a significant difference at the 0.05 level. Human Relations showed no significant difference, $t(445) = -1.105$, $p = 0.27$. For Performance Evaluation, $t(445) = -1.724$, $p = 0.086$, suggesting no significant gender differences. For Administrative and Technical Performance, $t(445) = -1.152$, $p = 0.25$, indicating no difference. Finally, the Total for Supervisory Performance revealed $t(445) = -1.658$, $p = 0.098$, indicating no significant gender-based differences. In conclusion, the only significant gender difference found in the study was in Professional Development, where females scored higher than males.

Table (5): results of independent sample two t-tests to examine the differences in Electronic Supervision and Supervisory Performance based on gender

variables	gender	N	Mean	Std. Deviation	t	Sig. (2-tailed)
Electronic Supervision Competencies	Male	271	3.37	0.78	0.777	0.438
	female	176	3.31	0.73		
Electronic Supervision Methods	Male	271	4.04	0.55	-1.23	0.22
	female	176	4.11	0.55		
Electronic Supervision Applications	Male	271	3.96	0.66	-0.092	0.927
	female	176	3.96	0.67		
Total for E-supervision	Male	271	3.74	0.60	-0.036	0.971
	female	176	3.74	0.58		
Supervisory Planning	Male	271	4.26	0.53	-1.187	0.236
	female	176	4.32	0.53		
Curriculum Development and Teaching Methods	Male	271	4.14	0.64	-0.65	0.516
	female	176	4.18	0.61		
Professional	Male	271	4.29	0.51	-2.322	0.021*

Development	female	176	4.40	0.53		
	Male	271	4.65	0.41		
Human Relations	female	176	4.69	0.41	-1.105	0.27
	Male	271	4.42	0.50		
Performance Evaluation	female	176	4.50	0.49	-1.724	0.086
	Male	271	4.44	0.48		
Administrative and Technical Performance	female	176	4.50	0.49	-1.152	0.25
	Male	271	4.37	0.42		
Total for Supervisory Performance	female	176	4.43	0.41	-1.658	0.098

* The difference is significant at the 0.05 level

Table (6) presents the results of a one-way ANOVA conducted to examine the differences in Electronic Supervision and Supervisory Performance based on years of experience. For Electronic Supervision Competencies, the results showed no significant differences between groups, $F(2, 444) = 0.115$, $p = 0.891$. Similarly, Electronic Supervision Methods revealed no significant differences, $F(2, 444) = 1.739$, $p = 0.177$. Electronic Supervision Applications also showed no significant differences, $F(2, 444) = 0.325$, $p = 0.722$. The Total for E-supervision also indicated no significant differences, $F(2, 444) = 0.341$, $p = 0.711$.

In the domain of Supervisory Planning, the analysis revealed no significant differences, $F(2, 444) = 0.052$, $p = 0.949$. For Curriculum Development and Teaching Methods, no significant differences were found, $F(2, 444) = 0.922$, $p = 0.398$. Similarly, Professional Development showed no significant differences, $F(2, 444) = 0.323$, $p = 0.724$. Human Relations also showed no significant differences, $F(2, 444) = 1.967$, $p = 0.141$. In the area of Performance Evaluation, the results indicated no significant differences, $F(2, 444) = 0.025$, $p = 0.976$. Administrative and Technical Performance showed no significant differences either, $F(2, 444) = 1.713$, $p = 0.181$. Finally, the Total for Supervisory Performance did not show

significant differences, $F(2, 444) = 0.461$, $p = 0.631$. In conclusion, the findings indicate that there are no significant differences in Electronic Supervision or Supervisory Performance based on years of experience.

Table (6): results of one-way ANOVA to examine the differences in Electronic Supervision and Supervisory Performance based on years of experience

Variables	Source of variance	Sum of Squares	df	Mean Square	F	Sig.
Electronic Supervision Competencies	Between Groups	0.134	2	0.067	0.115	0.891
	Within Groups	257.582	444	0.58		
	Total	257.716	446			
Electronic Supervision Methods	Between Groups	1.038	2	0.519	1.739	0.177
	Within Groups	132.531	444	0.298		
	Total	133.569	446			
Electronic Supervision Applications	Between Groups	0.289	2	0.145	0.325	0.722
	Within Groups	197.362	444	0.445		
	Total	197.651	446			
Total for E-supervision	Between Groups	0.242	2	0.121	0.341	0.711
	Within Groups	157.536	444	0.355		
	Total	157.777	446			
Supervisory Planning	Between Groups	0.029	2	0.015	0.052	0.949
	Within Groups	124.161	444	0.28		
	Total	124.191	446			
Curriculum Development and Teaching Methods	Between Groups	0.721	2	0.36	0.922	0.398
	Within Groups	173.574	444	0.391		
	Total	174.295	446			
Professional Development	Between Groups	0.176	2	0.088	0.323	0.724
	Within Groups	121.196	444	0.273		
	Total	121.373	446			
Human Relations	Between Groups	0.661	2	0.331	1.967	0.141
	Within Groups	74.631	444	0.168		
	Total	75.293	446			
Performance	Between Groups	0.012	2	0.006	0.025	0.976

Evaluation	Within Groups	110.665	444	0.249		
	Total	110.677	446			
Administrative and Technical Performance	Between Groups	0.813	2	0.406	1.713	0.181
	Within Groups	105.294	444	0.237		
	Total	106.106	446			
Total for Supervisory Performance	Between Groups	0.161	2	0.08	0.461	0.631
	Within Groups	77.335	444	0.174		
	Total	77.496	446			

This result is consistent with the findings of The study by Quzuq (2022)., which indicated there were no significant differences were found at ($\alpha \leq 0.05$) in the degree of educational supervision in public schools of the northern West Bank governorates as perceived by the educational supervisors according to the variables of (gender, qualification, specialization), whereas significant differences were found according to the variables of experience years and governorate. - There were no significant differences were found at ($\alpha \leq 0.05$) in the degree of electronic leadership and its obstacles in public schools of the northern West Bank governorates as perceived by the educational supervisors according to the variables of (gender, qualification, specialization, experience years, and governorate).

CONCLUSION

In conclusion, this study provides valuable insights into the current state of electronic supervision and its relationship with supervisory performance among supervisors in the Ministry of Education in the Sultanate of Oman. The results from Table (3) indicate that supervisors demonstrate a high level of competence in applying electronic methods and applications of supervision, although their overall electronic supervision competencies remain moderate. The findings highlight that while there is high confidence in using electronic supervision tools, there is still room for further development, particularly in enhancing overall competence in this area.

The correlation analysis, as shown in Table (4), reveals strong positive relationships between electronic supervision and various aspects of supervisory performance. This suggests that higher levels of competence in electronic supervision are linked to better supervisory performance, particularly in areas such as performance evaluation and supervisory planning. However, weaker correlations were observed in areas like human relations and curriculum development, indicating that there may be less of an impact of electronic supervision on these dimensions. This indicates potential areas for improvement in integrating electronic supervision tools to enhance overall performance in these areas.

Furthermore, gender-based differences were explored through independent samples t-tests (Table 5). The results revealed no significant differences between males and females in electronic supervision competencies or most dimensions of supervisory performance, with the only exception being professional development, where females scored significantly higher than males. This suggests that, overall, gender does not play a significant role in the effectiveness of electronic supervision or supervisory performance, except in the domain of professional development.

Lastly, the one-way ANOVA results (Table 6) indicate that years of experience do not significantly impact the supervisors' performance in electronic supervision or their overall supervisory performance. This suggests that the level of experience among supervisors does not necessarily correlate with better performance in either the electronic supervision tasks or the broader supervisory responsibilities.

Overall, the study indicates that while supervisors in Oman demonstrate a positive attitude and competence in electronic supervision, further training and development could enhance their overall effectiveness, especially in less impacted

areas such as human relations and curriculum development. Moreover, the lack of significant differences based on gender and experience highlights the need for uniform training and support systems that cater to all supervisors, irrespective of these factors.

RECOMMENDATIONS

Based on the findings of this study, the following recommendations are proposed:

1. **Enhance Training and Professional Development:** Given that supervisors show high competence in electronic supervision methods and applications but moderate overall competencies, it is recommended to implement targeted training programs that focus on improving electronic supervision competencies across all dimensions. This could include workshops, webinars, and hands-on training sessions to address gaps and build confidence in less experienced areas.
2. **Focus on Human Relations and Curriculum Development:** The results suggest weaker correlations between electronic supervision and human relations, as well as curriculum development and teaching methods. Therefore, it is recommended to develop strategies that integrate electronic supervision tools more effectively into these areas. Supervisors should receive additional training on how to use digital platforms to enhance communication, collaboration, and curriculum planning.
3. **Ongoing Support for Supervisors:** Continuous support through mentoring and peer networks could help supervisors navigate the challenges of electronic supervision. Establishing communities of practice or online forums where supervisors can share experiences and best practices may help improve the implementation of electronic supervision tools and enhance performance across all domains.

4. Gender-Neutral Development Programs: The study revealed no significant gender differences in most dimensions of electronic supervision, except for professional development. Thus, it is crucial to ensure that professional development opportunities are equally accessible to all supervisors, irrespective of gender, and are designed to address the specific needs and interests of both male and female supervisors.
5. Tailored Interventions Based on Experience Levels: The lack of significant differences based on years of experience in the study suggests that experience alone does not necessarily lead to better performance in electronic supervision. To address this, supervisors of all experience levels should have access to tailored interventions, ensuring that both new and experienced supervisors can improve their skills in electronic supervision.

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